

Vector Integration: Evaluating a Line Integral Along a Straight Line Path

1. Line Integral Parameterization

Name of Concept

Parametric Representation of a 3D Straight Line Path for Line Integration.

Core Mathematical Formulas

The symmetric equation of a straight line in 3D passing through points (x_1, y_1, z_1) and (x_2, y_2, z_2) equated to a scalar parameter t is:

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1} = t$$

The general differential line integral expansion equation is:

$$\int_C \vec{F} \cdot d\vec{r} = \int_C (f_1 dx + f_2 dy + f_3 dz)$$

Beginner-Friendly Explanation of Operations

- The Core Strategy (Parameterization):** A line integral requires us to sum values along a path where x , y , and z change simultaneously. Integrating with three different variables at once is difficult. To simplify, we express variables x , y , and z in terms of a single, shared dummy variable t (the parameter). Think of t as a timeline tracking your progress along the path: at the start line $t = 0$, and at the destination $t = 1$.
- Converting the Path Equations:** By using the 3D line formula, we divide the distance traveled along an axis by the total directional length of that axis. Equating these fractions to t allows us to extract explicit individual equations for coordinates: $x(t)$, $y(t)$, and $z(t)$.
- Converting the Differentials:** Because the coordinates change in terms of t , their microscopic steps (dx , dy , dz) must also be converted to terms of dt using standard differentiation. For example, if $x = 2t$, then taking the derivative gives $\frac{dx}{dt} = 2$, which means $dx = 2 dt$. This converts our triple-variable integration problem into a straightforward single-variable calculus integral with respect to t .

2. Step-by-Step Problem Solution

Question

Evaluate the line integral $\int_C \vec{F} \cdot d\vec{r}$ for the vector field $\vec{F} = 3x^2\vec{i} + (2xz - y)\vec{j} + z\vec{k}$ along the straight line path joining the origin $(0, 0, 0)$ to the point $(2, 1, 3)$.

Step 1: Identify components and map the endpoints

Extract the components of the vector field $\vec{F}$:

$$f_1 = 3x^2$$

$$f_2 = 2xz - y$$

$$f_3 = z$$

Assign the coordinate boundary points to the line equation variables:

$$(x_1, y_1, z_1) = (0, 0, 0)$$

$$(x_2, y_2, z_2) = (2, 1, 3)$$

Step 2: Set up the 3D straight line parametric template

Substitute the boundary point coordinates into the standard symmetric equation:

$$\frac{x-0}{2-0} = \frac{y-0}{1-0} = \frac{z-0}{3-0} = t$$
$$\frac{x}{2} = \frac{y}{1} = \frac{z}{3} = t$$

Step 3: Solve for x , y , and z explicitly in terms of t

Equate each individual fraction to the parameter t :

$$x = 2t$$

$$y = t$$

$$z = 3t$$

Step 4: Determine the integration limits for the parameter t

Find the interval of t that maps the path from start to end:

- At the starting point $(0, 0, 0)$: substituting $x = 0$ into $0 = 2t$ yields $t = 0$.
- At the ending point $(2, 1, 3)$: substituting $x = 2$ into $2 = 2t$ yields $t = 1$.

Thus, our integration boundaries are from $t = 0$ to $t = 1$.

Step 5: Compute the coordinate differentials in terms of dt

Take the derivative of each coordinate equation with respect to t :

$$dx = 2 dt$$

$$dy = 1 dt$$

$$dz = 3 dt$$

Step 6: Substitute the parametric equations into the vector components

Rewrite the field component expressions so they depend only on the variable t :

$$f_1 = 3(2t)^2 = 3(4t^2) = 12t^2$$

$$f_2 = 2(2t)(3t) - t = 12t^2 - t$$

$$f_3 = 3t$$

Step 7: Substitute all expressions into the line integral equation

Assemble the full integral equation by substituting the component functions and differential equations:

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 [(12t^2 \cdot 2 dt) + ((12t^2 - t) \cdot 1 dt) + (3t \cdot 3 dt)]$$

Factor out the common differential term dt and combine the terms inside the brackets:

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 [24t^2 + 12t^2 - t + 9t] dt$$

Combine like algebraic terms to find the final integrated function:

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 (36t^2 + 8t) dt$$

Step 8: Compute the single-variable definite integral

Apply standard calculus integration rules ($\int t^n dt = \frac{t^{n+1}}{n+1}$):

$$\int_C \vec{F} \cdot d\vec{r} = \left[36 \left(\frac{t^3}{3} \right) + 8 \left(\frac{t^2}{2} \right) \right]_0^1$$

$$\int_C \vec{F} \cdot d\vec{r} = [12t^3 + 4t^2]_0^1$$

Evaluate the expression by substituting the upper limit (1) and lower limit (0):

$$\int_C \vec{F} \cdot d\vec{r} = (12(1)^3 + 4(1)^2) - (12(0)^3 + 4(0)^2)$$

$$\int_C \vec{F} \cdot d\vec{r} = 12 + 4 = 16$$